Smart Water Distribution System with Time-Based Control and Intelligent Consumption Adjustment

Project Guide
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Abstract

This project, ***Smart Water Distribution System with Time-Based Control and Intelligent Consumption Adjustment***, aims to improve water management in urban areas using IoT and basic machine learning. Smart valve units (Sona walls) deliver water to households during fixed time slots, reducing wastage and ensuring fairness. A centralized dashboard enables **real-time monitoring**, while **machine learning** adjusts supply based on usage patterns and special events. The system supports sustainable development by promoting efficient, scalable, and adaptive water distribution.

Key features of the system include:

1. **Time-Based Water Allocation**: Each household receives water only during specific time slots, similar to mobile recharge validity, ensuring fair distribution.
2. **Machine Learning Integration**: The system intelligently adjusts water distribution schedules during weekends, festivals, or based on historical usage patterns.
3. **Smart Valve Control (Sona Wall):** Electronic valves controlled by Arno microcontrollers regulate water flow at each household level.
4. **Centralized Monitoring Dashboard**: A real-time web-based interface for the water department to monitor, analyze, and manage water distribution efficiently.
5. **Scalable and Data-Driven System**: Designed for scalability across communities or cities, with data analytics to support informed decision-making and reduce wastage.

SDG Goals:

1. **SDG 6 – Clean Water and Sanitation**
   Ensures availability and sustainable management of water by minimizing wastage, improving access, and promoting efficient use through smart time-based distribution.
2. **SDG 9 – Industry, Innovation, and Infrastructure**
   Supports the development of resilient infrastructure and fosters innovation through the integration of IoT and machine learning in public utility systems.
3. **SDG 11 – Sustainable Cities and Communities**
   Enhances urban resilience by implementing smart technologies that enable equitable and sustainable water distribution for growing city populations.
4. **SDG 12 – Responsible Consumption and Production**
   Encourages efficient use of resources and reduction of waste by monitoring and managing water usage intelligently.

Advantages of Proposed System:

- Fair and regulated water distribution to every household.
- Reduction in water wastage through automated control.
- Data-driven insights for authorities to optimize supply.
- Scalable architecture that can be extended to entire localities or cities.
- User-friendly dashboard for visualization and control.
- Machine learning integration enables adaptive scheduling without manual intervention.

Conclusion:

This project offers a practical and scalable solution to modern water distribution challenges. By combining IoT-based control systems, machine learning intelligence, and real-time dashboards, it ensures efficient, fair, and adaptive water delivery. The proposed model not only addresses the current shortcomings of traditional systems but also lays a foundation for future smart city applications in utility management.